Question	Answer	Marks
6(a)	potential difference per unit current	**B1**
6(b)(i)	$\rho = RA / L$	**C1**
	$= (0.33 \times 8.0 \times 10^{-6}) / 2.4$ $= 1.1 \times 10^{-6}\ \Omega\,\text{m}$	**A1**
6(b)(ii)	$Q = It$	**C1**
	$= 4.7 \times 5.0 \times 60$ $= 1400\ \text{C}$	**A1**
6(b)(iii)	$I = nAvq$ $n = 4.7 / (8.0 \times 10^{-6} \times 0.16 \times 10^{-3} \times 1.60 \times 10^{-19})$	**C1**
	$= 2.3 \times 10^{28}\ \text{m}^{-3}$	**A1**
6(c)(i)	correct symbols for resistor and thermistor, shown correctly connected in series with the battery	**B1**
6(c)(ii)	(as temperature increases) resistance of thermistor decreases	**M1**
	(total resistance decreases so) greater current (in circuit/wire) so power (dissipated in the wire) increases **or** (total resistance decreases so) greater (share of) p.d. across wire so power (dissipated in the wire) increases	**A1**

Question	Answer	Marks